EECE 306 Lab 1 notebook. Building the Toolbox, Constants and Vector Algebra

Team 3 Mitch Nazareth. TODO replace with your team number and member names.

Fill in every item marked TODO, then run `publish('Lab01_notebook.m')` from the toolbox root and print the resulting HTML to PDF. Three to five pages. The notebook is graded on required results, verification evidence, interpretation, honesty about problems, and presentation.

Contents

Setup

This cell must run clean from the toolbox root.

```
clear; close all;
disp(version)
```

```
11.3.0
```

Part I. Physical constants

The two derived constants must be computed, not typed in.

```
fprintf('eps0 = %.10e F/m\n', em.const.eps0());
fprintf('mu0  = %.10e H/m\n', em.const.mu0());
fprintf('c0   = %.10e m/s\n', em.const.c0());
fprintf('eta0 = %.6f ohm\n',  em.const.eta0());
em.test.assertClose(em.const.c0(), 1/sqrt(em.const.mu0()*em.const.eps0()), 1e-14, 'c0 derived');
em.test.assertClose(em.const.eta0(), sqrt(em.const.mu0()/em.const.eps0()), 1e-14, 'eta0 derived');
disp('Derived constant checks passed.')
```

```
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0   = 2.9979245808e+08 m/s
eta0 = 376.730314 ohm
Derived constant checks passed.
```

Part II. Vector algebra on the required checks

```
fprintf('mag([3 4 0])  = %.15g   (expect 5 exactly)\n', em.vec.mag([3 4 0]));
disp('unit([3 4 0]) ='); disp(em.vec.unit([3 4 0]))
disp('unit([0 0 0]) (expect all NaN) ='); disp(em.vec.unit([0 0 0]))
fprintf('angle perpendicular = %.15g  (expect pi/2)\n', em.vec.angle([1 0 0],[0 1 0]));
ap = em.vec.angle([1 0 0],[-1 0 0]);
fprintf('angle antiparallel  = %.15g, real = %d  (expect pi, 1)\n', ap, isreal(ap));
disp('fromTo([1 1 1],[2 3 4]) ='); disp(em.vec.fromTo([1 1 1],[2 3 4]))
```

```
mag([3 4 0])  = 5   (expect 5 exactly)
unit([3 4 0]) =
   0.6000   0.8000        0
unit([0 0 0]) (expect all NaN) =
   NaN   NaN   NaN
angle perpendicular = 1.5707963267949  (expect pi/2)
angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
   1   2   3
```

Part II continued. Shape discipline

Every function must accept Nx3 input and return the documented shape.

```
A = randn(100,3); B = randn(100,3);
fprintf('mag    input 100x3 output %s\n', mat2str(size(em.vec.mag(A))));
fprintf('unit   input 100x3 output %s\n', mat2str(size(em.vec.unit(A))));
fprintf('angle  input 100x3 output %s\n', mat2str(size(em.vec.angle(A,B))));
fprintf('fromTo input 100x3 output %s\n', mat2str(size(em.vec.fromTo(A,B))));
```

```
mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
```

Interpretation

TODO three to six sentences in your own words. Why must c0 and eta0 be computed from eps0 and mu0 rather than typed in as numbers, and why is == the wrong way to compare a computed field value against a formula.

c0 and eta0 are derrived from esp0 and mu0, and hardcoding these values introduces roudning errors that would be prevelent in later computations. Aditionally, it is easier to work with values we know are correct, rather than continuously changes fixed constants rounded to a point where we would reduce rounding errors. This makes the code easier to work with in the long run.

Problems encountered

TODO state plainly anything that did not work and what was tried. An honest account of a failure earns more credit than silence. Write NONE if the session was clean.

When I started working on these, I felt very confused aobut the file structure, where to find the starter files, and such. I know that this would have been much easier had I joined sooner. Aditionally, the team I joined had confusion about the structure of the submission, so we all had to sort of go back and fix some of the work. We have since got up and running on github, and things have now been much better. We made us all seperate branches, and we can push our work to main when it is deemed good enough to work. I think having our work on github will really bring us a long way this semester.

- Mitch

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
--------------------------------------------------------------
11.3.0
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0   = 2.9979245808e+08 m/s
eta0 = 376.730314 ohm
Derived constant checks passed.
mag([3 4 0])  = 5   (expect 5 exactly)
unit([3 4 0]) =
   0.6000   0.8000        0
unit([0 0 0]) (expect all NaN) =
   NaN   NaN   NaN
angle perpendicular = 1.5707963267949  (expect pi/2)
angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
   1   2   3
mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
  PASS  test_lab01            0.01 s
--------------------------------------------------------------
  1 passed, 0 failed, 1 total
  All tests passing.
```

Published with GNU Octave 11.3.0